KUMARAGURU COLLEGE OF TECHNOLOGY

Department of Artificial Intelligence and Data Science

DATA SCIENCE AND VISUALIZATION (24ADI204)

WEEK 1 REPORT

Team Formation & Data Hunting

Team 16: Youth Smoking and Drug Consumption Analysis

DATA-VIZ HACK-A-BIT 2026–2027

Team Member Name	**Roll Number**
Lakshmi Prabha M M	25BAD049

1. Week 1 Contribution:

Researched statistical analysis techniques and Exploratory Data Analysis (EDA) requirements suitable for the project.

2. Work Done:

During Week 1, my individual work focused on researching statistical techniques and Exploratory Data Analysis requirements for the Youth Smoking and Drug Consumption Analysis project. I explored methods that can be used to understand the distribution of youth smoking and drug consumption-related variables and identify meaningful patterns and relationships within the dataset.

3. Topics Researched:

- **Descriptive Statistics –** Researched measures of central tendency, including Mean, Median and Mode, to summarize numerical features and understand the general characteristics of the dataset.
- **Measures of Dispersion –** Studied Variance and Standard Deviation to understand the spread and variation in numerical values related to drug consumption.
- **Skewness and Kurtosis –** Explored how skewness identifies the asymmetry of data distributions and how kurtosis describes the heaviness of distribution tails.

- **Univariate Analysis –** Researched techniques for analysing individual variables using Histograms, Boxplots and Frequency Distributions.
- **Bivariate Analysis –** Studied methods to examine relationships between two variables using Scatter Plots and other suitable visualizations.
- **Correlation Analysis –** Explored how correlation helps identify positive and negative relationships between numerical features.
- **Data Visualization –** Researched Matplotlib and Seaborn libraries for creating statistical visualizations to understand patterns and distributions in the dataset.

4. Outcome of My Contribution:

This research helped me understand the importance of statistical analysis and EDA in identifying meaningful patterns in youth smoking and drug consumption data. I gained knowledge of descriptive statistical measures and visualization techniques that can be applied during the upcoming stages of the project. This research provided a foundation for performing statistical analysis and identifying relationships between variables in the selected dataset.

5. Reference:

Project notebook: DSV Team 16 – Youth Smoking and Drug Consumption Analysis.